Assignment: Accessibility Audit and Fixes

Week: 10 **Points:** 2 points	**Due Date:** Monday, March 30, 2026 **Submit:** ZIP file (HTML, CSS, report)

Assignment Overview

Conduct an accessibility audit on a webpage, identify issues, and fix them to meet WCAG 2.1 AA standards. You will learn to use accessibility testing tools and implement fixes that make websites usable for everyone, including people with disabilities. This is a critical skill for professional web designers!

Requirements

Choose ONE Option

Fix Your Own Work (Recommended)

- Use your Project 1
- Audit and fix accessibility issues

IMPORTANT: Running a Local Server

Browser extensions like WAVE will NOT work if you just open your HTML file directly. You need to run a local server first.

How to Run a Local Server in VS Code:

1. **Install the Live Server extension (if you haven't already):**
   - Click the Extensions icon in VS Code (or press Ctrl+Shift+X)
   - Search for "Live Server"
   - Click "Install"
2. **Start your local server:**
   - Open your project folder in VS Code
   - Right-click on your index.html file
   - Select "Open with Live Server"
   - Your page will automatically open in your browser at http://localhost:5500

3. **Now you can test with WAVE and Lighthouse:**
   - WAVE extension will work because your page is on localhost
   - Lighthouse will work in Chrome DevTools (F12 then Lighthouse tab)

> **Tip:** Keep Live Server running while you test and make fixes. When you save changes in VS Code, the page will automatically refresh in your browser!

Part 1: Accessibility Audit (Required)

Test Your Page Using:

- WAVE Tool - Browser extension (requires localhost URL)
- Lighthouse - Built into Chrome DevTools (F12 then Lighthouse tab)
- Keyboard Navigation - Unplug your mouse and try navigating with Tab key
- Color Contrast Checker - WebAIM at webaim.org/resources/contrastchecker

Document at Least 5 Issues:

- What the issue is
- Why it is a problem
- WCAG guideline it violates (if applicable)
- How you plan to fix it

Part 2: Implement Fixes (Required)

Must Fix Issues in These Categories:

1. Semantic HTML (1 to 2 fixes)

- Replace with semantic elements where appropriate
- Ensure proper heading hierarchy (no skipped levels)
- Use semantic elements correctly

2. Alt Text & Media (1 to 2 fixes)

- Add or improve alt text on images
- Ensure decorative images have alt=""
- Add captions or transcripts if using video/audio

3. Keyboard Navigation (1 to 2 fixes)

- Ensure all interactive elements are keyboard accessible
- Add visible focus states (:focus styling)
- Check tab order makes sense
- Skip navigation link (optional but encouraged)

4. Color Contrast (1 to 2 fixes)

- Ensure text meets 4.5:1 contrast ratio (AA standard)
- Fix any contrast issues with buttons, links, or text

- Consider colorblind-friendly palette

5. ARIA & Screen Readers (1 to 2 fixes)

- Add appropriate ARIA labels where needed
- Add aria-label to icon-only buttons
- Use role attributes if necessary
- Add aria-hidden="true" to decorative icons

Part 3: Accessibility Report (Required)

Create a document (PDF, Markdown, or Word) that includes:

Before Section:

- Screenshot of WAVE report (showing errors)
- Screenshot of Lighthouse accessibility score
- List of 5+ issues found with descriptions

After Section:

- Screenshot of improved WAVE report
- Screenshot of improved Lighthouse score
- List of fixes implemented
- Code snippets showing before/after

Reflection (3 to 5 sentences):

- What surprised you?
- What was most challenging?
- How will this change your approach to web design?

File Requirements

What to Submit:

```
LastName_A11y/
    index.html (fixed version)
    style.css  (fixed version)
    accessibility-report.pdf (or .md, .docx)
    screenshots/
        before-wave.png
        after-wave.png
        before-lighthouse.png
        after-lighthouse.png
```

File Naming:

`LastName_A11y.zip`
Example: `Kraishan_A11y.zip`

Grading Rubric (2 points total)

Criteria	Points	What I'm Looking For
Audit Documentation	**0.5**	Identifies 5+ accessibility issues. Clear descriptions of problems. Includes before screenshots from WAVE and Lighthouse.
Semantic HTML Fixes	**0.3**	Proper use of semantic elements. Correct heading hierarchy. Meaningful structure for screen readers.
Alt Text & Media	**0.2**	All images have appropriate alt text. Decorative images use alt="". Meaningful descriptions.
Keyboard Navigation	**0.3**	All interactive elements keyboard accessible. Visible focus states. Logical tab order.
Color Contrast	**0.3**	Text meets 4.5:1 ratio. Links distinguishable. Good contrast on all elements.
ARIA & Accessibility	**0.2**	Appropriate use of ARIA labels and roles. Helps screen reader users. Not over-used or misused.
Report & Reflection	**0.2**	Complete before/after documentation. Screenshots included. Thoughtful reflection.

Common Accessibility Issues & Fixes

Issue 1: Missing Alt Text

Problem:

```
<img src="coffee.jpg">
```

Fix:

```
<img src="coffee.jpg"
     alt="Freshly brewed espresso in a white cup">
```

Decorative images:

```
<img src="decorative.png" alt="">
```

Issue 2: Non-Semantic HTML

Problem:

```
<div class="header">
    <div class="navigation">...</div>
</div>
```

Fix:

```
<header>
    <nav>...</nav>
</header>
```

Issue 3: Poor Color Contrast

Problem:

```
.text {
    color: #999999; /* Light gray */
    background: #FFFFFF; /* White */
    /* Contrast ratio: 2.85:1 - FAILS */
}
```

Fix:

```
.text {
    color: #666666; /* Darker gray */
    background: #FFFFFF; /* White */
    /* Contrast ratio: 5.74:1 - PASSES */
}
```

Issue 4: No Focus States

Problem:

```
a {
    outline: none; /* BAD! */
}
```

Fix:

```
a:focus {
```

```
    outline: 2px solid #007bff;
    outline-offset: 2px;
}
```

Issue 5: Skipped Heading Levels

Problem:

```
<h1>Page Title</h1>
<h3>Section</h3>  <!-- Skipped h2! -->
```

Fix:

```
<h1>Page Title</h1>
<h2>Section</h2>  <!-- Proper hierarchy -->
```

Issue 6: Icon-Only Button

Problem:

```
<button>
    <i class="icon-menu"></i>
</button>
```

Fix:

```
<button aria-label="Open navigation menu">
    <i class="icon-menu" aria-hidden="true"></i>
</button>
```

Tips for Success

Semantic HTML:

- `Use <header>, <nav>, <main>, <footer>`
- `Use <button> for buttons, not <div>`
- Keep heading hierarchy (h1 to h2 to h3, no skipping)

Alt Text:

- Describe what the image shows, not "image of..."
- Be specific: "Golden retriever puppy" not "dog"
- `Decorative images: Use alt=""`
- Keep it concise (under 150 characters)

Color & Contrast:

- Test with color contrast checker
- Do not rely on color alone to convey information
- Use patterns or text in addition to color

Focus States:

- Never remove outline without replacing it
- Make focus states obvious (2 to 3px outline)
- Test by tabbing through your page

ARIA:

- Use semantic HTML first (better than ARIA)
- Only add ARIA when semantic HTML is not enough
- Do not over-use ARIA attributes
- `aria-label for icon-only buttons`

Helpful Resources

Testing Tools:

- WAVE Extension at wave.webaim.org/extension
- Lighthouse in Chrome DevTools (F12)
- WebAIM Contrast Checker at webaim.org/resources/contrastchecker

Learning Resources:

- WCAG 2.1 Guidelines at w3.org/WAI/WCAG21/quickref
- WebAIM Articles at webaim.org/articles

Need Help?

Office Hours: Tuesday & Thursday 2:00 PM to 3:30 PM
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